



TEXAS A&M UNIVERSITY

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Industrial & Systems Engineering

# Project LiAlSon

Spring 2026 Capstone Report

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# Capstone Final Report

for

Team ML8: Project LiAlSon

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Submitted to fulfill the requirement for a final report in  
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## Executive Summary

This project was undertaken to develop an automated anomaly-detection workflow for fused sonar–LiDAR point cloud data to improve the safety, speed, and cost-effectiveness of underwater infrastructure inspections for MerLion Advisory. To address the limitations of traditional diver-based inspections, the team (1) generated controlled synthetic underwater environments with embedded anomalies using open-source 3D modeling tools, (2) designed a standardized preprocessing and labeling pipeline for fused point cloud data, and (3) implemented and evaluated a machine-learning-based object-level detection and classification approach using PointNet.

The final system demonstrated strong performance on synthetic data, achieving approximately 80% accuracy in binary anomaly detection and F1 scores exceeding 0.80 for all anomaly classes. The automated workflow enables consistent, objective identification of underwater anomalies while significantly reducing operational risk. Economic analysis indicates that adopting the proposed solution can reduce daily inspection expenses by over 50% and increase inspection throughput by approximately 80% compared to manual diver-based methods.

Based on these results, it is recommended that MerLion use this system as a proof-of-concept detection engine and pursue limited retraining with real sonar–LiDAR data to transition toward operational deployment. With incremental refinement and integration, the workflow developed in Project LiAISon provides a scalable foundation for modern, automated underwater inspection operations.

# Table of Contents

|                                                                        |     |
|------------------------------------------------------------------------|-----|
| Executive Summary .....                                                | iii |
| Table of Contents .....                                                | iv  |
| List of Figures .....                                                  | vi  |
| List of Tables .....                                                   | vi  |
| 1. Phase 1: Discover .....                                             | 7   |
| 1.1 Sponsor Overview .....                                             | 7   |
| 1.2 Problem Overview .....                                             | 7   |
| 1.3 Problem Statement .....                                            | 7   |
| 1.4 Client Needs .....                                                 | 8   |
| 1.5 Project Charter .....                                              | 8   |
| 1.6 Literature Review .....                                            | 12  |
| 2. Phase 2: Define .....                                               | 12  |
| 2.1 As-Is Concept of Operations .....                                  | 12  |
| 2.2 System Requirements .....                                          | 13  |
| 2.3 Engineering Actions .....                                          | 13  |
| 2.4 Needs-Requirements-Actions-Technical Objectives Traceability ..... | 14  |
| 3. Phase 3: Develop .....                                              | 15  |
| 3.1 To-Be Concept of Operations .....                                  | 15  |
| 3.2 Solution Concepts .....                                            | 15  |
| 3.3 Analysis of Alternatives .....                                     | 16  |
| 3.4 Selected Solution .....                                            | 17  |
| 4. Phase 4: Deliver .....                                              | 17  |
| 4.1 Final Design .....                                                 | 17  |
| 4.1.1 Design Architecture .....                                        | 18  |
| 4.1.2 Design Implementation .....                                      | 18  |
| 4.1.3 Requirements Traceability .....                                  | 20  |
| 4.2 Cost Analysis .....                                                | 20  |
| 4.3 Performance Analysis .....                                         | 21  |
| 4.4 Risk Analysis .....                                                | 23  |
| 4.5 Constraints and Engineering Standards .....                        | 23  |

|       |                                                                   |    |
|-------|-------------------------------------------------------------------|----|
| 4.5.1 | Client Constraints.....                                           | 23 |
| 4.5.2 | Engineering Standards .....                                       | 23 |
| 5.    | Conclusions and Recommendations .....                             | 24 |
| 5.1   | Conclusions and Recommendations .....                             | 24 |
| 5.2   | Future Work .....                                                 | 25 |
| 6.    | References .....                                                  | 25 |
| 7.    | Appendix A: Supporting Data & Documentation .....                 | 26 |
| 7.1   | Reference Pictures from Boat Demo .....                           | 26 |
| 7.2   | Example Scenes .....                                              | 26 |
| 8.    | Appendix B: ABET Documentation.....                               | 27 |
| 8.1   | ABET Criterion 3: Student Outcomes .....                          | 27 |
| 8.1.1 | Student Outcome 1 .....                                           | 27 |
| 8.1.2 | Student Outcome 2.....                                            | 29 |
| 8.1.3 | Student Outcome 3.....                                            | 30 |
| 8.1.4 | Student Outcome 4.....                                            | 33 |
| 8.1.5 | Student Outcome 5.....                                            | 35 |
| 8.1.6 | Student Outcome 6.....                                            | 38 |
| 8.1.7 | Student Outcome 7.....                                            | 39 |
| 8.2   | ABET Criterion 5d: Curriculum Capstone Requirements .....         | 40 |
| 8.2.1 | Engineering Standards .....                                       | 40 |
| 8.2.2 | Multiple Realistic Constraints.....                               | 40 |
| 8.2.3 | Use of Knowledge and Skills Acquired in Earlier Course Work ..... | 41 |
|       | Final Deliverables Checklist .....                                | 42 |

## List of Figures

|                                                                        |    |
|------------------------------------------------------------------------|----|
| Figure 1. Final design overview.....                                   | 18 |
| Figure 2. Confusion matrix for Phase 1: Binary classification.....     | 22 |
| Figure 3. Confusion matrix for Stage 2: Multiclass classification..... | 22 |
| Figure 4. MerLion's boat collecting data .....                         | 26 |
| Figure 5. MerLion's boat labeled .....                                 | 26 |
| Figure 6. Example synthetic scene in Blender with anomalies .....      | 26 |
| Figure 7. Ground truth versus predicted anomaly placement .....        | 27 |

## List of Tables

|                                                                                                 |    |
|-------------------------------------------------------------------------------------------------|----|
| Table 1. Needs-Requirements-Actions-Technical Objectives Traceability Matrix.....               | 14 |
| Table 2. Economic analysis comparing manual diving to proposed solution.....                    | 21 |
| Table 3. Success metrics for PointNet model for binary anomaly detection.....                   | 21 |
| Table 4. Success metrics for PointNet model for multiclass classification anomaly detection.... | 22 |
| Table 5. Previous DAEN Courses Referenced during Capstone Project.....                          | 41 |

# 1. Phase 1: Discover

The purpose of Phase 1 was to set the direction for our project. Phase 1 was all about exploring the project and getting everyone familiar with the problem at hand. The way to achieve that was to gather as much information as possible. Defining the problem, identifying client needs, and coming up with technical objectives were all a big part of Phase 1.

## 1.1 Sponsor Overview

MerLion Advisory is our sponsor. They are a provider of maritime domain awareness and response services in seaports, coastal and inland waters. They focus on protecting and sustaining resources through a fusion of high-performance skills with emerging technologies to achieve client objectives. Their current focus revolves around revolutionizing the way the waters are monitored and secured. Their mission is unparalleled visibility above and beneath water, to reduce the need for human personnel to ensure the safety and security of the Maritime industry.

## 1.2 Problem Overview

The current problem is that underwater inspections of piers, ports, and coastal infrastructure rely heavily on manual diver-based methods. This poses a few challenges which include but are not limited to: the process being extremely time consuming and hazardous; the data collected through that method is at many times inconsistent or incomplete due to poor visibility, currents, and equipment limitations. This is where the goal of this project comes in.

## 1.3 Problem Statement

Our problem statement is: Detecting stationary anomalies in synthetic fused sonar–LiDAR point clouds to support safer underwater infrastructure inspections. Our focus is data generation, preprocessing, anomaly definition, detection workflow, visualization, and evaluation. All datasets will be synthetically generated through open-source 3D modeling tools like Blender/Blender.

The original problem revolved around combining underwater and above water data to get a single dataset which would aid MerLion in detecting anomalies. However, due to limitations in accessing the datasets, the focus was shifted to just detecting anomalies using synthetic datasets that were generated by the team.

## 1.4 Client Needs

The client's needs can be broken down into three points. The first one being that the client needs a safer, faster, and more accurate underwater inspection process reducing risk and cost. The second is that the client needs an automated anomaly detection workflow applied to fused sonar-LiDAR point clouds generated with open-source 3D modeling tools. Lastly, there is also a need for a validation report and reproducible, documented workflow.

## 1.5 Project Charter

|                        |                                                                                   |                        |                                                                                                                                                    |
|------------------------|-----------------------------------------------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Team</b>    | Maddie Bird<br>Leisha Khapre<br>Inayaa Khoja<br>Gaurav Shatoor<br>Jade Winebright | <b>Project Sponsor</b> | MerLion Advisory<br>Arnie Tyler<br>Email:<br><a href="mailto:at Tyler@MerLionAdvisory.com">at Tyler@MerLionAdvisory.com</a><br>Phone: 972-807-3659 |
| <b>Faculty Advisor</b> | Dr. Alfredo Garcia<br>Anshul Yadav                                                |                        |                                                                                                                                                    |

### 1. Project Title and Overview

**Project Title:** Project LiAISon

Project LiAISon focuses on detecting stationary anomalies in fused sonar–LiDAR point clouds to support maritime safety and underwater infrastructure monitoring. All point clouds used in this project are synthetically generated through open-source 3D modeling tools (e.g., Blender or BIAnder for sensor simulation), allowing controlled simulation of underwater environments and irregular structures such as debris, deformities, or unnatural formations.

A separate team is responsible for producing fused sonar–LiDAR data. Our scope begins once the fused point clouds are provided, enabling us to focus on developing preprocessing workflows, implementing anomaly detection methods, and evaluating detection results using synthetic ground-truth scenarios. By narrowing the scope to detection rather than data fusion, the project aims to prototype techniques that can inform future automated inspection systems and reduce reliance on manual diver-based assessments.

### 2. Problem Statement

Underwater inspections of piers, ports, and coastal infrastructure currently rely heavily on manual diver-based methods. These inspections are time-consuming, hazardous, and often yield inconsistent or incomplete data due to environmental limitations such as poor visibility, currents, and equipment constraints. As a result, organizations face increased operational risk, delayed maintenance decisions, and higher costs for repairs.

By developing an anomaly detection pipeline, this project seeks to improve inspection speed, accuracy, and operational safety.

### 3. Project Goals and Objectives

**Project Goal:** Develop an anomaly detection workflow for fused sonar–LiDAR point clouds generated using open-source 3D modeling tools.

**Objectives:**

- Synthetic Dataset Creation
  - Generate synthetic underwater environments in Blender or BIAnder and export fused point clouds in standard formats such as .ply or .pcd.
  - Include embedded ground-truth anomalies (e.g., debris, structural irregularities) to support evaluation.
- Preprocessing Pipeline
  - Clean and normalize fused point clouds using noise filtering, downsampling, and alignment.
  - Standardize data formats and metadata to ensure the workflow is reproducible.
- Anomaly Definition and Labeling
  - Define anomaly categories relevant to underwater infrastructure
  - Create ground-truth labels for the synthetic datasets.
- Detection Algorithm Development
  - Implement and compare multiple anomaly-detection approaches, such as machine learning models, statistical outlier detection, and geometric feature-based methods.
- Visualization Tools
  - Develop 3D visualization outputs that highlight detected anomalies through color coding or overlays.
- Evaluation and Validation
  - Evaluate detection performance using precision, recall, F1 score, false-positive and false-negative rates, and runtime efficiency.
  - Validate results against the ground-truth anomalies embedded in the synthetic datasets.

### 4. Stakeholders and Roles

**Sponsor - MerLion Advisory:** Provides project direction.

**Team - ML-ATE:** Responsible for developing and implementing the project solution.

- Leisha Khapre - Leader / Quality Assurance / Reviewer
- Maddie Bird - Project Management / Analyst / Documenter
- Inayaa Khoja - Project Management / Analyst / Documenter
- Jade Winebright - Documenter / Reviewer / Developer
- Gaurav Shatoor - Developer / Designer / Researcher

**Team Communication:** Weekly meetings, shared Google Drive; will use GitHub for version control.

**Instructor - Dr. Abuabara:** Guides milestones and deliverables.

**Advisors - Dr. Garcia, Anshul Yadav:** Provide technical guidance, review deliverables, and offer feedback.

## 5. Timeline and Major Milestones

- **Aug – Nov 2025:**  
Initial project exploration and development of charter drafts.
- **Dec 2025:**  
Finalize and present the completed project charter.
- **Dec 2025 – Jan 2026:**  
Create synthetic underwater scenes and generate sonar–LiDAR datasets. Begin preprocessing and standardizing fused point cloud formats.
- **Feb – Mar 2026:**  
Define anomaly categories and develop prototype detection algorithms. Conduct initial tests and refine preprocessing workflows as needed.
- **Mar – Apr 2026:**  
Evaluate detection performance using ground-truth anomalies and defined metrics. Refine algorithms based on results. Potentially produce 3D visualizations of detected anomalies and complete system integration.
- **May 2026:**  
Finalize documentation, validation summary, and deliver the final report and sponsor presentation.

## 6. Deliverables

- Synthetic sonar–LiDAR datasets generated from 3D underwater scenes, including raw and fused point clouds in standard formats (e.g., .ply, .pcd).
- Preprocessed and labeled point cloud datasets with documented anomaly ground truth.
- Anomaly detection scripts or notebooks implementing all tested algorithms, along with a reproducible code repository and environment files.
- 3D visualizations of raw, preprocessed, and anomaly-highlighted point clouds.
- Validation report summarizing performance metrics (precision, recall, F1 score, false-positive rate, runtime).
- Final documentation describing system design, methods, dataset-generation process, assumptions, and future improvements.

## 7. Success Criteria and Metrics

- Data Preparation
  - Point cloud data cleaned, organized, and prepared for analysis.
- Anomaly Definitions
  - Clear categories of stationary anomalies with corresponding ground-truth labels in synthetic data.
- Workflow Functionality
  - A complete pipeline that runs from input point clouds to anomaly outputs.
- Detection Performance

- Algorithms perform above baseline methods using metrics such as precision, recall, F1 score, and false-positive rate (no fixed thresholds defined yet).
- Visualization
  - Visual tools clearly display detected anomalies.
- Documentation & Reproducibility
  - The entire workflow is fully documented and reproducible.

## 8. Assumptions and Risks

### Assumptions

- Software used for synthetic data provides sufficient fully fused point clouds.
- Resolution is sufficient for anomaly detection.
- The workflow will still be valuable to MerLion although the project uses synthetic data.

### Risks

| Risk                                                  | Likelihood | Category | Owner            | Mitigation                                                                                    |
|-------------------------------------------------------|------------|----------|------------------|-----------------------------------------------------------------------------------------------|
| File loss                                             | 2          | Medium   | All Team Members | Using git commits often and backing up code                                                   |
| Scheduling Risks                                      | 3          | High     | Project Managers | Maintaining and updating timeline, conducting phase reviews                                   |
| Synthetic ground truth may not match real conditions. | 4          | Medium   | Developers       | Document uncertainty levels, and design synthetic scenarios that mimic real-world conditions. |
| Misinterpretation of outputs                          | 3          | Low      | Project Managers | Using non-technical summaries, regular meetings with stakeholders                             |
| Computational load                                    | 3          | Medium   | Team Members     | Mitigate via downsampling or cloud computing                                                  |

## 9. Ethical, Legal Standards and Constraints

The team will follow university standards and handle all project data securely and responsibly. All simulated datasets used for testing will be clearly labeled. All work will adhere to the principles of transparency, accountability, and reproducibility.

## 1.6 Literature Review

Similar problems have been identified in the past, and papers have been written outlining the solutions that were used by people to solve them. The most helpful paper that was found was about underwater 3D object detection from sonar point clouds (Shaukat et al., 2025). Their approach compared a deep learning method (SASA neural network trained on synthetic sonar data) with a traditional model-based template matching method. The findings of this approach can be found in Appendix A. Model-based methods proved more robust in noisy, cluttered underwater environments, while deep learning excelled only in controlled synthetic settings. Since we are using a synthetic dataset, it makes more sense to use the SASA model. Keeping that in mind, the limitations of this approach will be well documented.

Upon talking with our advisor, Dr. Garcia, it was also decided that we would try using another model- autoencoder- to compare the results and choose the better model out of the two. This is a neural network-based model which works by decreasing the dimensions of the dataset, which will be extremely helpful in this case since the data is high dimensional.

## 2. Phase 2: Define

After exploring the problem space during Phase 1, the focus of Phase 2 was to define the system and outline how the project will operate. While Phase 1 concentrated on understanding the sponsor's problem and gathering information, Phase 2 translated those findings into a structured operational model. This phase established the system requirements, engineering actions, and operational workflow needed to build the anomaly detection pipeline. It also clarified how data will be generated, processed, analyzed, and evaluated throughout the project.

### 2.1 As-Is Concept of Operations

The current operational concept focuses on detecting stationary anomalies within synthetic fused sonar-LiDAR point cloud data to support safer underwater infrastructure inspections. The system begins with the generation of synthetic underwater scenes using Blender and BIAnder, where anomalies are intentionally embedded into the environment. These scenes are converted into fused point clouds that simulate sonar and LiDAR scans.

Once the data is generated, it enters a Python-based processing pipeline. In this stage, the raw point cloud data is preprocessed through noise filtering and downsampling to reduce computational complexity while maintaining important structural features. The processed data is then passed into models, such as SASA.

After anomaly detection is completed, the results are evaluated using performance metrics such as precision, recall, and F1 score. These outputs ultimately produce actionable insights that can support maritime inspection decision-making.

The operational environment is structured around a centralized development team with sponsor oversight. All datasets are synthetic, and development emphasizes reproducibility and documentation. The system relies on open-source tools including Blender for scene creation, BLAInder for sonar–LiDAR simulation, Python for the processing pipeline and modeling, and GitHub for storage and version control.

The overall data lifecycle follows several key stages. First, underwater scenes are generated with embedded anomalies. Next, the point clouds are preprocessed through filtering and downsampling. The fused point clouds are exported in standardized formats such as PLY, PCD, or LAS. Detection models are then executed to identify anomalies, followed by validation using performance metrics. Finally, visualization overlays allow users to inspect the detected anomalies within the 3D point cloud environment.

## 2.2 System Requirements

The system requirements define the key capabilities the project must deliver:

- R1 – Data Generation: Generate synthetic underwater scenes containing labeled stationary anomalies.
- R2 – Preprocessing: Filter noise from raw point cloud data and downsample the data to reduce computational load.
- R3 – Detection: Identify stationary anomalies within the processed point cloud data and output predictions.
- R4 – Evaluation and Reproducibility: Evaluate detection results against ground truth labels and provide documented instructions to allow the pipeline to be reproduced and executed consistently.

## 2.3 Engineering Actions

To meet the defined system requirements, several engineering actions were identified.

- EA1 – Develop Synthetic Underwater Datasets: Create approximately three underwater scenes with embedded anomalies and generate fused sonar–LiDAR point clouds using Blender.
- EA2 – Preprocessing Implementation: Develop the preprocessing pipeline to perform noise filtering, downsampling, and standardization of the point cloud data.
- EA3 – Label Anomalies and Define Schema: Define categories of stationary anomalies and establish a consistent labeling schema for the dataset.

- EA4 – Develop and Compare Detection Models: Implement anomaly detection models and compare their performance to determine the most effective approach.
- EA5 – Build Validation Framework: Create a framework to evaluate detection results using metrics such as precision, recall, and F1 score and generate documentation summarizing the results.

## 2.4 Needs-Requirements-Actions-Technical Objectives Traceability

The traceability matrix connects the client's needs to the technical work performed by the team. This ensures every engineering action directly contributes to solving the sponsor's problem.

Table 1. Needs-Requirements-Actions-Technical Objectives Traceability Matrix

| Client Needs                                    | System Requirements                                    | Technical Objectives                                                                                               | Engineering Actions                                                                                                                                          |
|-------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>N1:</b> Better underwater inspection process | <b>R1:</b> Data Generation<br><b>R2:</b> Preprocessing | <b>TO1:</b> Develop point clouds<br><b>TO2:</b> Build preprocessing pipeline<br><b>TO3:</b> Define/label anomalies | <b>EA1:</b> Develop synthetic underwater datasets (~3 scenes) with Blender<br><b>EA2:</b> Preprocessing implementation<br><b>EA3:</b> Label Anomaly & Schema |
| <b>N2:</b> Automated anomaly detection          | <b>R3:</b> Detection                                   | <b>TO4:</b> Implement anomaly detection models,<br><b>TO5:</b> Create 3D visualization tools for detecting         | <b>EA3:</b> Label Anomaly & Schema<br><b>EA4:</b> Develop & compare detection models' performance                                                            |
| <b>N3:</b> Reproducible, documented workflow    | <b>R4:</b> Evaluation & Reproducibility                | <b>TO6:</b> Validation report, final documentation                                                                 | <b>EA5:</b> Build validation framework to evaluate performance & document                                                                                    |

## 3. Phase 3: Develop

Phase 3 focused on generating, evaluating, and refining solution concepts that could meet the system requirements defined in Phase 2. While earlier phases established the problem space and operational framework, Phase 3 concentrated on building the synthetic data pipeline, developing candidate anomaly-detection approaches, and comparing their feasibility. This phase also produced the To-Be Concept of Operations, which outlines how MerLion's real sonar/LiDAR data will eventually flow through the detection pipeline. By the end of Phase 3, the team had generated multiple detection concepts, evaluated their tradeoffs, and identified the most promising direction for final implementation.

### 3.1 To-Be Concept of Operations

The To-Be system represents the future operational workflow once MerLion integrates automated anomaly detection into its maritime inspection process. In this envisioned system, MerLion's boat-mounted sonar and LiDAR sensors collect underwater data during inspection missions. This raw sensor data is transformed into point cloud formats such as .ply or .xyz, which are then ingested into the Python based preprocessing pipeline.

The pipeline preprocesses the data, applies noise filtering and downsampling, and prepares the point cloud for model evaluation. The anomaly detection model performs two tasks: (1) binary detection of whether an anomaly is present, and (2) classification of the anomaly type (e.g., cube, sphere, boat). The system outputs both a 0/1 anomaly flag and a labeled anomaly class, along with visual overlays for operator review.

This To-Be workflow increases automation, reduces reliance on manual diver inspections, and provides consistent, objective detection results. It also supports MerLion's long-term goal of integrating real-time anomaly detection into their maritime safety operations.

### 3.2 Solution Concepts

During Phase 3, the team explored three primary solution concepts:

#### Concept 1: Scene-Level Anomaly Detection

- **Brainstorming:** Simple binary “anomaly or not” idea for full 50×50 meter scene
- **Initial Screening:** Highly feasible; minimal labeling; strong CONOPS alignment
- **Detailed Development:**
  - Models: SASA, Autoencoder
  - Output: 0/1 anomaly flag for entire scene
- **Synthesis:** Fully supports baseline requirements; no major gaps

#### Concept 2: Object-Level Detection

- **Brainstorming:** Not only detect anomalies but identify them
- **Initial Screening:** More complex but feasible; adds interpretability

- Detailed Development:
  - Models: SASA or similar,
  - Output: 0/1 anomaly flag for entire scene anomaly classification
- Synthesis: Fills traceability gaps for localization requirements

### **Concept 3: CNN Image-Based Segmentation**

- **Brainstorming:** Familiar 2D approach based on prior coursework
- **Initial Screening:** Feasible; partially aligned with 3D CONOPS but useful as a supplement
- **Detailed Development:**
  - Models: CNN
  - Output: 2D bounding boxes or masks
- **Synthesis:** Adds visualization and operator clarity; complements 3D methods

## **3.3 Analysis of Alternatives**

The three solution concepts were compared across economic cost, technical risk, and performance potential:

- Scene-Level Detection
  - Lowest cost and simplest labeling
  - Low technical risk
  - Good detection performance but no localization
- Object-Level Detection & Localization
  - Higher labeling and compute cost
  - Moderate technical risk
  - Strongest alignment with MerLion's needs due to interpretability and localization
- CNN Image-Based Segmentation
  - Moderate cost
  - Medium risk due to 2D limitations
  - Good visualization but misaligned with 3D objectives

Object-level detection emerged as the most balanced and operationally relevant approach, despite its higher complexity.

### 3.4 Selected Solution

The selected solution is Object-Level Detection and Classification implemented using PointNet. This approach best satisfies MerLion's needs for anomaly identification, interpretability, and future integration with real sonar and LiDAR data. PointNet was chosen because it can process raw, unordered point clouds directly while remaining lightweight and computationally efficient. It also outperformed DGCNN in overall accuracy and training speed, while supporting both binary detection and multiclass anomaly classification. In addition, the approach aligns well with the To-Be CONOPS and MerLion's operational constraints, providing a strong foundation for a scalable, real-world underwater anomaly-detection system.

## 4. Phase 4: Deliver

Phase 4 focused on finalizing the system, validating performance, and preparing the deliverables for MerLion. This included completing the end-to-end pipeline, conducting economic and risk analyses, refining visualizations, and producing documentation for integration with real sonar/LiDAR data.

### 4.1 Final Design

The final design consists of a full end-to-end pipeline that begins with synthetic underwater scene generation in Blender and ends with anomaly detection, classification, and visualization. The system processes raw point clouds, applies standardized preprocessing, and uses trained machine-learning models to identify stationary anomalies such as cubes, spheres, and boats.

#### 4.1.1 Design Architecture

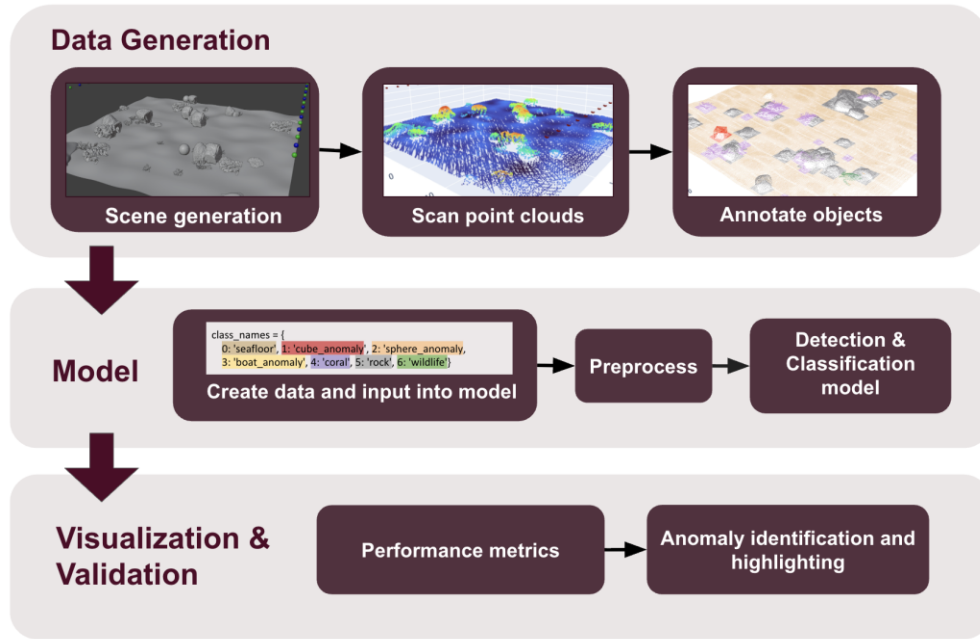


Figure 1. Final design overview

To create the data, underwater scenes are created in Blender with embedded anomalies. 120 scenes were generated with random object placement and are scanned to produce point clouds with ground-truth labels. Point clouds then undergo downsampling and noise filtering to ensure consistent quality.

For anomaly detection and classification, a PointNet model was selected due to its ability to process unordered point clouds efficiently and its strong performance across anomaly classes. Detected anomalies are highlighted, and performance is evaluated using precision, recall, F1 score, and confusion matrices. This architecture mirrors MerLion’s future workflow, where real sonar/LiDAR data will replace synthetic inputs.

#### 4.1.2 Design Implementation

To deploy this system with real-world data, the sponsor will need a Python environment running Python 3.10 or later, with the following libraries installed: PyTorch (version 2.0 or later with CUDA 11.8+ for GPU acceleration), Open3D, scikit-learn, pandas, numpy, matplotlib, plotly, and tqdm, all installable via pip. A CUDA-capable GPU is strongly recommended- the system was developed and tested on an NVIDIA T4.

The pipeline expects point cloud data in CSV format with four mandatory columns: x, y, z, and class, where x, y, z are metric Cartesian coordinates and class is an integer label. Since the sponsor likely operates a combined LiDAR and sonar sensor suite rather than LiDAR alone, a preprocessing step will be required before data enters the pipeline: the LiDAR and sonar returns

must be registered into a single unified coordinate frame using a known sensor extrinsic calibration (rigid body transform between the two sensors), after which the merged point cloud should be exported as a single CSV per scan.

If real-world data arrives unlabeled- which is the expected case at deployment- the class column should be set to a placeholder value of -1 for all points, and the system will run in inference-only mode, skipping all ground truth comparisons and outputting only predicted labels, cluster centroids, and confidence scores. Seafloor points should be removed prior to ingestion either by the sponsor's existing bathymetric segmentation pipeline or by a depth threshold filter, since the model was trained to operate exclusively on non-seafloor returns.

Once data is in the correct format, the sponsor loads the two trained model weights- `stage1_binary.pt` and `stage2_classifier.pt`- alongside the `pipeline_config.json` file which contains all hyperparameters, then calls the `run_pipeline` function which internally performs DBSCAN spatial clustering on the input points, passes each cluster through Stage 1 (anomaly vs. not anomaly) and routes positives to Stage 2 (cube, sphere, or boat classification), and returns a results list with a predicted label, confidence score, cluster centroid in world coordinates, and bounding box extents for every detected object. These outputs can be directly ingested into the sponsor's existing GIS or mission planning software as georeferenced object detections, or visualized in-session using the built-in Plotly 3D viewer.

If the sponsor's operational data differs substantially in point density, noise characteristics, or object scale from the synthetic training scenes, fine-tuning both models on a small set of labelled real-world examples- even 10 to 20 labelled scenes- is strongly advised before operational use, as domain shift between synthetic and real sonar/LiDAR data is the most likely source of performance degradation in deployment.

### 4.1.3 Requirements Traceability

| Client Needs                                    | System Requirements                                    | Technical Objectives                                                                                               | Engineering Actions                                                                                                                              | Final Design                                                                  |
|-------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>N1:</b> Better underwater inspection process | <b>R1:</b> Data Generation<br><b>R2:</b> Preprocessing | <b>TO1:</b> Develop point clouds<br><b>TO2:</b> Build preprocessing pipeline<br><b>TO3:</b> Define/label anomalies | <b>EA1:</b> Develop synthetic underwater datasets with Blender<br><b>EA2:</b> Preprocessing implementation<br><b>EA3:</b> Label Anomaly & Schema | Generated scenes with labeled anomalies and scanned to generate point clouds. |
| <b>N2:</b> Automated anomaly detection          | <b>R3:</b> Detection                                   | <b>TO4:</b> Implement anomaly detection models,<br><b>TO5:</b> Create 3D visualization tools for detecting         | <b>EA3:</b> Label Anomaly & Schema<br><b>EA4:</b> Develop & compare detection models' performance                                                | Developed 3 detection models; compared using F1 score, Recall and Accuracy    |
| <b>N3:</b> Reproducible, documented workflow    | <b>R4:</b> Evaluation & Reproducibility                | <b>TO6:</b> Validation report, final documentation                                                                 | <b>EA5:</b> Build validation framework to evaluate performance & document                                                                        | Wrote reproducible code; documented and stored in Github                      |

## 4.2 Cost Analysis

The synthetic pipeline eliminates over \$20k in startup field-trial costs and boosts inspection throughput by 80% by removing manual processing bottlenecks. Analyst workload shifts from time-consuming searching to quick verification, while autonomous USV deployment reduces diver hours and associated insurance liability. Because the workflow is fully reproducible, MerLion can scale to new ports with no additional overhead.

Table 2. Economic analysis comparing manual diving to proposed solution

| Metric          | Manual Diver Inspection                     | Project LiAlSon (USV + ML)            |
|-----------------|---------------------------------------------|---------------------------------------|
| Daily Cost      | \$2,500 – \$5,000+                          | \$600 – \$1,200 (Operating/Rental)    |
| Personnel       | 3-5 people (Diver, Tender, Supervisor)      | 1 Operator                            |
| Safety Risk     | High (Drowning, decompression, murky water) | Zero (Human stays on shore/deck)      |
| Time to Results | Days                                        | Real-Time                             |
| Coverage Rate   | Limited by diver fatigue & air supply       | Continuous (Battery/Fuel dependent)   |
| Data Quality    | Subjective (Diver's visual/tactile notes)   | Objective (High-res Sonar/LiDAR logs) |

### 4.3 Performance Analysis

Performance of the binary and multiclass models were evaluated using precision, recall, F1 score, and confusion matrices. For binary detection, PointNet achieved 0.80 accuracy (Table 3). For multiclass classification, PointNet performed strongly on cube and sphere anomalies, with F1 scores above 0.85, and moderate performance on boat anomalies (Table 4). PointNet provided the best balance of accuracy, speed, and computational efficiency.

Table 3. Success metrics for PointNet model for binary anomaly detection

|             | Precision | Recall | F1 Score | Accuracy |
|-------------|-----------|--------|----------|----------|
| Not Anomaly | 0.93      | 0.76   | 0.84     | 0.80     |
| Anomaly     | 0.64      | 0.89   | 0.75     |          |

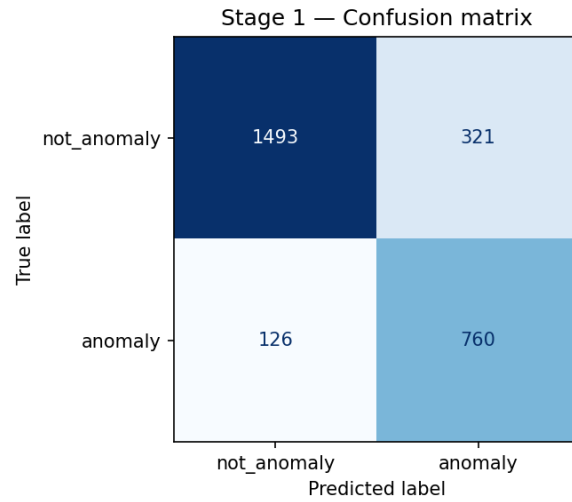


Figure 2. Confusion matrix for Phase 1: Binary classification

Table 4. Success metrics for PointNet model for multiclass classification anomaly detection

|        | Precision | Recall | F1 Score | Accuracy |
|--------|-----------|--------|----------|----------|
| Cube   | 0.97      | 0.81   | 0.89     | 0.91     |
| Sphere | 0.97      | 0.99   | 0.90     |          |
| Boat   | 0.70      | 0.96   | 0.81     |          |

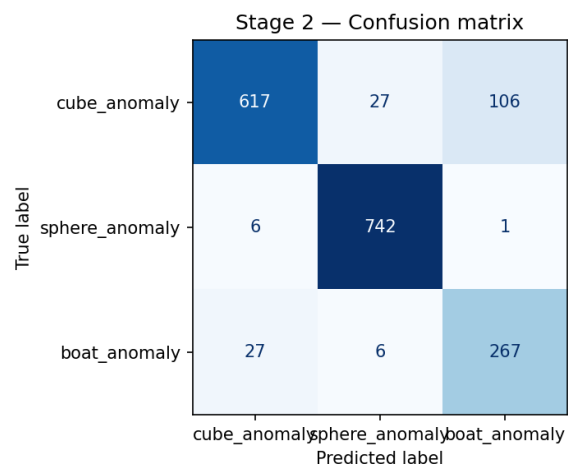


Figure 3. Confusion matrix for Stage 2: Multiclass classification

## 4.4 Risk Analysis

The major risks in this project included the synthetic–real gap, since synthetic underwater scenes may not fully capture real environmental complexity; this was mitigated by diversifying scene conditions and planning to retrain models once real sonar/LiDAR data becomes available. Human error in scene setup and labeling also posed a risk, addressed through standardized instructions and version-controlled workflows. Finally, repeatability was a concern due to variation in scene parameters, which was mitigated by enforcing a fixed preprocessing pipeline and consistent labeling schema.

## 4.5 Constraints and Engineering Standards

Our design was guided by the sponsor’s operational constraints and the engineering standards identified in the project charter. These constraints shaped our technical decisions, model selection, and workflow structure to ensure the final system was feasible, reproducible, and aligned with MerLion’s needs.

### 4.5.1 Client Constraints

Our solution meets all of MerLion’s constraints by operating entirely within limited computational, financial, and regulatory boundaries. The team worked under restricted access to high-performance computing, requiring lightweight models and efficient preprocessing. Model accuracy limitations were addressed through iterative testing and evaluation across multiple architectures.

Time constraints were managed through a modular pipeline and standardized workflows. Economic constraints were met by relying exclusively on free tools such as Google Colab, Blender, and open-source Python libraries, all of which operate within memory and runtime limits. Legal and ethical constraints were also central to the design. The team ensured compliance with data privacy and usage guidelines and avoided the use of proprietary or sensitive data.

### 4.5.2 Engineering Standards

The final design adheres to all engineering standards identified in the project charter and reinforced during Phase 4.

#### ISO 19115: Geographic Information Metadata

This standard guided our approach to documenting spatial datasets. We applied it by maintaining consistent metadata for point clouds, including scene parameters, anomaly labels, and coordinate frames.

#### ISO/IEC 25012: Data Quality Model

This standard shaped our requirements for accuracy, consistency, and completeness. Our preprocessing pipeline and labeling schema were built to ensure high-quality, standardized training data.

#### PEP 8: Python Code Style Standard

All preprocessing, detection, and evaluation scripts follow PEP 8 to ensure readability, maintainability, and long-term usability for MerLion’s engineering team.

These standards ensured that the system was not only functional but also maintainable, safe, and aligned with industry expectations.

## 5. Conclusions and Recommendations

Phase 4 brought together all components of the project into a complete, validated anomaly-detection pipeline ready for MerLion’s future integration with real data. This phase finalized the system architecture, evaluated model performance, documented reproducibility, and demonstrated how the workflow can reduce cost, improve safety, and streamline underwater inspection operations.

### 5.1 Conclusions and Recommendations

This project successfully developed and validated an end-to-end anomaly-detection pipeline for synthetic fused sonar–LiDAR point cloud data in support of safer and more efficient underwater infrastructure inspections. The final system meets the primary sponsor needs by providing an automated, reproducible workflow that reduces reliance on hazardous manual diver inspections while improving inspection speed and consistency.

Using open-source tools and synthetically generated data, the team designed a complete pipeline that includes scene generation, preprocessing, anomaly detection, classification, evaluation, and visualization. The selected PointNet-based object-level detection approach demonstrated strong performance, achieving approximately 80% accuracy for binary anomaly detection and F1 scores above 0.80 for all anomaly classes in multiclass classification. These results validate the feasibility of applying machine-learning-based detection to fused point cloud data and provide a strong technical foundation for future deployment with real sonar–LiDAR inputs.

From an operational and economic perspective, the proposed solution delivers significant value to the sponsor. By shifting inspections from diver-based methods to autonomous sensing and automated analysis, the workflow has the potential to reduce daily inspection costs by more than 50% and increase inspection throughput by approximately 80%. Most importantly, it substantially improves safety by keeping personnel out of hazardous underwater environments while still delivering high-quality, objective inspection data.

The team recommends that MerLion treat this system as a proof-of-concept detection engine that can be incrementally integrated into their operational pipeline. With some adaptation- primarily retraining or fine-tuning using a small set of labeled real-world scans- the solution can transition from synthetic testing to real deployments. The documented, modular design ensures that MerLion can scale, maintain, and extend the workflow as sensor capabilities and inspection requirements evolve.

## 5.2 Future Work

While this project demonstrates the feasibility and value of automated anomaly detection using synthetic fused point cloud data, several opportunities exist for future teams to build upon this work.

The most impactful next step is domain adaptation using real sonar–LiDAR data. Future teams should collaborate closely with MerLion to collect and label a limited set of real inspection scans and fine-tune the existing PointNet models. This step would directly address the synthetic-to-real domain gap identified as a primary project risk and significantly improve deployment readiness.

Additional future work should focus on expanding anomaly classes and environmental complexity. The current system detects a small, well-defined set of stationary anomalies. Future efforts could introduce more realistic infrastructure features such as corrosion patterns and partially buried debris, improving the system’s operational relevance.

Overall, Project LiAISon establishes a strong technical and methodological foundation. With targeted extensions focused on real data integration, model robustness, and operational deployment, future capstone teams can continue advancing this work toward a fully operational underwater inspection solution.

## 6. References

Shaukat, M. S., Käckenmeister, Y., Bader, S., & Kirste, T. (2025). *Towards training-free underwater 3D object detection from sonar point clouds: A comparison of traditional and deep learning approaches* (Version 1). arXiv. <https://doi.org/10.48550/arXiv.2508.18293>

## 7. Appendix A: Supporting Data & Documentation

### 7.1 Reference Pictures from Boat Demo



Figure 4. MerLion's boat collecting data



Figure 5. MerLion's boat labeled

### 7.2 Example Scenes

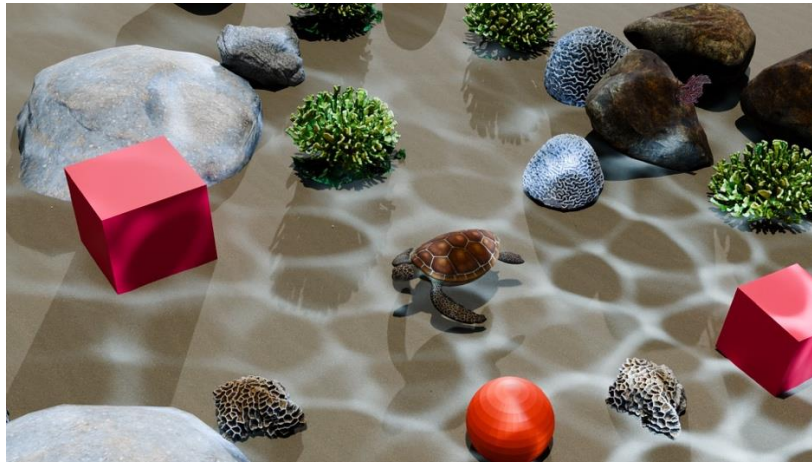


Figure 6. Example synthetic scene in Blender with anomalies

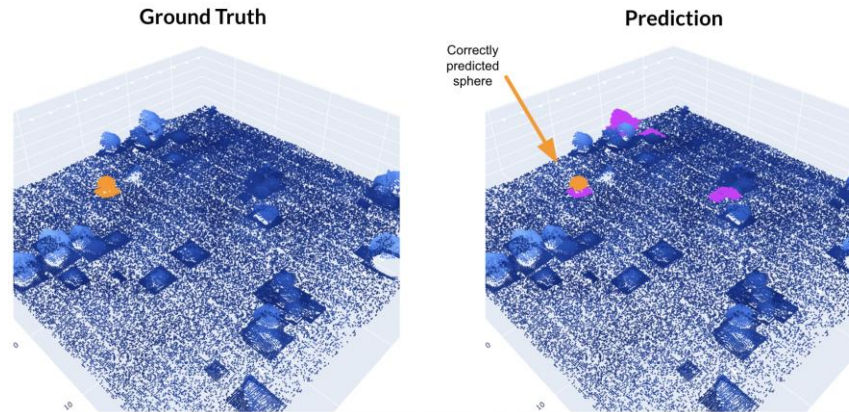


Figure 7. Ground truth versus predicted anomaly placement

## 8. Appendix B: ABET Documentation

### 8.1 ABET Criterion 3: Student Outcomes

#### 8.1.1 Student Outcome 1

*An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.*

**Question 1:** Was your senior design problem a “complex engineering problem<sup>1</sup>”?

Yes. Our project required integrating synthetic sonar-LiDAR data generation, high-density point-cloud preprocessing, and machine-learning-based anomaly detection while meeting MerLion’s constraints on safety, accuracy, and reproducibility. The problem involved multiple stakeholders and challenging requirements, such as restricted access to real sonar-LiDAR data, limited

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<sup>1</sup> **Complex engineering problems** include one or more of the following characteristics: involving wide-ranging or conflicting technical issues, having no obvious solution, addressing problems not encompassed by current standards and codes, involving diverse groups of stakeholders, including many component parts or sub-problems, involving multiple disciplines, or having significant consequences in a range of contexts.

computing resources, and the need for a fully documented, repeatable workflow. See *1.4. Client Needs* on pg. 8

**Question 2:** Was your proposed solution based on the principles of engineering?

Yes. Our solution applied core engineering principles by designing a complete system architecture that includes data generation, preprocessing, model development, and validation. We used structured engineering actions (EA1-EA5) to ensure the workflow met the system requirements for automation, safety, and reproducibility. See *2.2 System Requirements* on pg. 13.

**Question 3:** Was your proposed solution based on principles of science?

Yes. The project relied on realistic simulation of underwater systematic experimentation to evaluate how well different detection models performed. We used controlled synthetic scenes, consistent preprocessing steps, and measurable performance metrics to ensure our results were scientifically valid and repeatable. See *3.3 Analysis of Alternatives* on pg. 16.

**Question 4:** Was your proposed solution based on principles of mathematics?

Yes. Mathematical concepts played an important role in our solution because our detection models rely on numerical patterns in 3D point-cloud data, and the neural-network architecture of PointNet uses mathematical operations to learn those patterns. We also used mathematical performance metrics, such as precision, recall, F1 score, and confusion matrices, to evaluate how well each model detected anomalies. See *4.3 Performance Analysis* on pg. 21.

### 8.1.2 Student Outcome 2

*An ability to apply engineering design<sup>2</sup> to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.*

**Question 1:** How did you consider “public health” in your proposed solution?

Our solution indirectly supports public health by reducing the need for divers to operate in hazardous underwater environments, which lowers the risk of injuries that require medical intervention. By shifting inspections to automated sonar/LiDAR-based detection, the system helps prevent accidents that could strain emergency or medical resources in coastal areas. See *1.2 Problem Overview* on pg. 7.

**Question 2:** How did you consider “public safety” in your proposed solution?

Public safety was a central design consideration because underwater structural failures, such as damaged piers or submerged debris, pose risks to workers, vessels, and port infrastructure. Our anomaly-detection workflow enables earlier identification of hazards, reducing the likelihood of accidents during maritime operations. See *1.2 Problem Overview* on pg. 7.

**Question 3:** How did you consider “public welfare” in your proposed solution?

Our solution supports public welfare by improving the reliability and safety of maritime infrastructure, which directly affects coastal communities that depend on ports for employment, transportation, and economic activity. By enabling earlier detection of underwater hazards, the system reduces the likelihood of infrastructure failures that could disrupt commerce or endanger workers. See *1.2 Problem Overview* on pg. 7.

**Question 4:** How did you consider “global” issues/factors in your proposed solution?

Ports worldwide face similar safety, maintenance, and cost pressures. By developing a reproducible, open-source anomaly-detection workflow, our solution can potentially be adapted globally to improve maritime safety and reduce reliance on hazardous diver-based inspections. See *3.1 To-Be Concept of Operations* on pg. 15.

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<sup>2</sup> **Engineering design** is a process of devising a system, component, or process to meet desired needs and specifications within constraints. It is an iterative, creative, decision-making process in which the basic sciences, mathematics, and engineering sciences are applied to convert resources into solutions. Engineering design involves identifying opportunities, developing requirements, performing analysis and synthesis, generating multiple solutions, evaluating solutions against requirements, considering risks, and making trade-offs, to obtain a high-quality solution under the given circumstances. For illustrative purposes only, examples of possible constraints include accessibility, aesthetics, codes, constructability, cost, ergonomics, extensibility, functionality, interoperability, legal considerations, maintainability, manufacturability, marketability, policy, regulations, schedule, standards, sustainability, or usability.

**Question 5:** How did you consider “cultural” issues/factors in your proposed solution?

Cultural factors relevant to our project include decision-making standards. Our workflow provides transparent model results that can be reviewed, validated, and approved at any level. Attitudes toward work and safety vary across ports and agencies, with some cultures emphasizing traditional diver expertise and others prioritizing automation to reduce risk. Our system aligns with both by enhancing safety without removing human oversight. Finally, many coastal communities place strong cultural value on environmental stewardship, and our use of synthetic data and reduced diver disturbance supports these priorities by minimizing ecological impact. See *1.4. Client Needs* on pg. 8.

**Question 6:** How did you consider “social” issues/factors in your proposed solution?

Socially, the project reduces risk for divers, supports safer working conditions, and promotes long-term job stability by shifting workers toward supervisory and analytical roles rather than hazardous underwater tasks. This aligns with MerLion’s mission to reduce human exposure to dangerous environments. See *1.4. Client Needs* on pg. 8.

**Question 7:** How did you consider “environmental” issues/factors in your proposed solution?

Our solution considers environmental factors by reducing the need for repeated diver-based inspections, which can disturb marine ecosystems. By relying on automated detection, the workflow minimizes water disturbance. This approach supports environmentally responsible inspection practices while maintaining data quality. See *1.2 Problem Overview* on pg. 7.

**Question 8:** How did you consider “economic” issues/factors in your proposed solution?

Economically, the system reduces inspection costs by eliminating over startup field-trail expenses and lowering daily operational costs through automation. By reducing diving hours and shifting analysts from manually searching to quick verification, the workflow increases throughput by 80% and decreases labor and insurance costs. This creates a positive economic impact by improving efficiency, lowering operational expenses, and enabling scalable deployment. See *4.2 Cost Analysis* on pg. 20.

### 8.1.3 Student Outcome 3

*An ability to communicate effectively with a range of audiences.*

**Question 1:** How did you communicate with your project sponsors?

We communicated with our sponsor, Arnie Tyler, primarily through email and scheduled Zoom meetings, typically as needed or during major project milestones. Email was used to coordinate meeting times, clarify expectations, and share progress updates. Meetings were used to review

deliverables, confirm technical direction, and ensure alignment with MerLion's operational needs.

Sample Sponsor Communication (Outgoing Email):

*"Hi Arnie,*

*Would you be free next week for a third meeting with all the groups? We have a couple questions that we would like to be clarified.*

*We'd be ok with having it on Monday or Wednesday 10:20–11:10 am CST, but if that doesn't work for you we can figure out another time!*

*Best Regards, "*

This sample illustrates our standard communication approach and how we coordinated and scheduled meetings throughout the project.

**Question 2:** How did you communicate with your project advisors?

We communicated with our faculty advisor, Dr. Alfredo Garcia, and our graduate advisor, Anshul Yadav, through email, weekly check-ins, and in-person meetings. Communication frequency varied from weekly to monthly depending on project phase. Advisors provided technical guidance, feedback on modeling approaches, and direction on next steps.

Sample Advisor Communication (Outgoing Email):

*"Dr. Garcia,*

*I hope you're doing well! Our DAEN capstone team wanted to reconnect and see if you might be available to meet next week to discuss our progress and next steps for the project.*

*A time that works for all of the group members is Tuesday (2/24) between 12:45 PM and 3:30 PM. If a time within that window doesn't fit your schedule, we'd be happy to coordinate based on your availability.*

*We're looking forward to catching up and getting your guidance as we move forward!*

*Thank you, "*

Sample Graduate Advisor Communication (Outgoing Email):

*"Hi Anshul!*

*We were thinking of maybe meeting next week. We met up with Dr. Alfredo Garcia today and he asked us to research more about the technology and come up with toy datasets, so maybe this week we can work on that and then meet next Friday (10/30) around 9 am to discuss this with you. Would that be alright with you?*

*Best, "*

These samples reflect how our communication style was clear, respectful, and focused on technical progress.

**Question 3:** In addition to the two parties stated above, who were other groups you communicated with during the course of this project?

In addition to our sponsor and advisors, we communicated regularly with other DAEN capstone teams, especially those working with the same sponsor. These communications occurred through shared class sessions and group meetings to coordinate questions, align expectations, and ensure consistent understanding of MerLion's needs.

We also frequently communicated with our three course professors: Dr. Alexander Abuabara, Mr. Jose Vazquez, Dr. Jon Elizondo. This communication occurred during class sessions and phase presentations where we received guidance on deliverables, timelines, and technical expectations.

#### 8.1.4 Student Outcome 4

*An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.*

**Question 1:** What were your team's ethical and professional responsibilities while you were working on developing solutions for your design project?

Our team's ethical and professional responsibilities centered on ensuring that our work upheld the NSPE principles of honesty, accuracy, and protection of public welfare. In alignment with NSPE Fundamental Canon 1, we designed our anomaly-detection workflow to enhance safety by reducing diver risk and minimizing human error. We also followed Canon 3, ensuring that all technical statements, results, and performance metrics were objective, truthful, and supported by evidence. We maintained integrity and transparency consistent with NSPE Professional Obligation 1, acknowledging limitations and avoiding misrepresentation. See [\*NSPE code of ethics: \(I\) Fundamental Canons 1; \(I\) Fundamental Canons 4; \(III\) Professional Obligations 1.\*](#)

**Question 2:** What were your team's professional and ethical responsibilities to the project sponsor?

Our responsibilities to MerLion were guided by NSPE Canon 4, which requires engineers to act as faithful agents for their clients. We communicated clearly, avoided conflicts of interest, and ensured that all deliverables were accurate, reproducible, and aligned with the sponsor's operational needs. Consistent with NSPE Rule 1(c), we protected all non-public information and avoided sharing any proprietary details. We also upheld Canon 6, conducting ourselves responsibly and professionally in all sponsor interactions. See [\*NSPE code of ethics: \(I\) Fundamental Canons 4; \(I\) Fundamental Canons 6; \(II\) Rules of Practice 1\(c\).\*](#)

**Question 3:** How did you consider and assess the global impact of your proposed solutions?

We considered the global impact of our solution by recognizing that underwater infrastructure inspection is a widespread challenge affecting ports, coastal communities, and maritime safety. In line with NSPE Professional Obligation 2(d), our approach improves how inspection data is examined. MerLion will continue gathering data as usual, but our model adds value by automatically identifying anomalies, supporting safer and more efficient inspections. Because it can be applied across different locations, it has the potential to improve inspection practices worldwide. See [\*NSPE code of ethics: \(III\) Professional Obligations 2\(d\).\*](#)

**Question 4:** How did you consider and assess the economic impact of your proposed solutions?

Our economic assessment followed NSPE Canon 1, which emphasizes protecting public welfare through efficient and responsible engineering solutions. By reducing startup field-trial costs and daily inspection expenses, our workflow provides a positive economic impact for ports and maritime agencies. The automation of anomaly detection increases throughput by 80%, reduces labor costs, and supports more cost-effective infrastructure maintenance. This aligns with NSPE

Professional Obligation 2(a), which encourages engineers to advance community well-being. See [\*NSPE code of ethics: \(I\) Fundamental Canons 1; \(III\) Professional Obligations 2\(a\)\*](#).

**Question 5:** How did you consider and assess the environmental impact of your proposed solutions?

Environmental considerations were guided by NSPE Professional Obligation 2(d), which emphasizes sustainability and environmental protection. Our approach reduces the need for repeated diver deployments, helping limit disturbance to marine ecosystems, seafloor habitats, and wildlife. We used synthetic data for development, while MerLion will rely on existing sonar and LiDAR systems, both of which avoid direct environmental harm. By enabling more efficient, autonomous inspections, the solution can also reduce vessel activity and fuel use, supporting more environmentally responsible maritime operations. See [\*NSPE code of ethics: \(III\) Professional Obligations 2\(d\)\*](#).

**Question 6:** How did you consider and assess the societal impact of your proposed solutions?

We evaluated societal impacts by considering how safer, more reliable underwater inspections benefit communities that depend on bridges, ports, pipelines, and coastal infrastructure. Consistent with NSPE Canon 1, our system enhances public safety by reducing diver risk and improving anomaly detection accuracy. The reproducible workflow also supports the broader public interest, aligning with NSPE Professional Obligation 2(a), which encourages engineers to advance the safety, health, and well-being of their communities. See [\*NSPE code of ethics: \(I\) Fundamental Canons 1; \(III\) Professional Obligations 2\(a\)\*](#).

### 8.1.5 Student Outcome 5

*An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.*

**Question 1:** How would you rate the effectiveness of your team and its members?

Our team was very effective in completing the project objectives. We structured the team into two main groups, scene generation and model testing, where one group would lead while the other supported, and then switch depending on the phase of the project. This made sure progress was always happening across both areas. In addition to that, we also had smaller individual responsibilities such as documentation, preprocessing, and evaluation, which helped keep everything organized and ensured that no part of the project was overlooked. Each member consistently showed up to meetings, completed their assigned tasks on time, and supported teammates when needed, which played a big role in keeping the project on track.

**Question 2:** How did you and your team members collectively provide leadership?

Leadership was shared across the team rather than assigned to one person. Each group took the lead in their area when needed, such as the scene generation team leading dataset creation and the model team leading detection and evaluation. At the same time, decisions like model selection, preprocessing methods, and evaluation metrics were made together as a group. This balance allowed us to stay organized while still making sure everyone contributed to key decisions.

**Question 3:** How did your team members create a collaborative and inclusive environment?

Our team created a collaborative environment by keeping communication open and consistent throughout the project. Our main form of communication was iMessage group chat, which made it easy to ask quick questions, share updates, and stay connected outside of meetings. We also held regular meetings and used shared tools like Google Drive and GitHub so everyone had access to the same information. Team members were comfortable asking questions, helping each other, and giving input on decisions, which made it easier to work through challenges as a group.

## Project Collaboration Agreement

**Course:** DAEN 459 Capstone Senior Design Planning – Spring 2026

**Project Title:** MerLion Advisory

**Team Name:** ML8

**Date:** 1/18/26

### 1. Purpose

This agreement outlines how team members will work together to successfully complete our capstone project, ensuring clear communication, accountability, and mutual respect.

### 2. Team Mission and Values

Our mission is to work collaboratively to deliver a high-quality and effective solution for the MerLion Advisory project while meeting all course requirements professionally and on time. Our core values are responsibility, accountability, transparency, respect, and collaboration.

Success means completing a strong, well-organized project, maintaining positive team collaboration, and gaining practical skills and experience.

### 3. Team Members & Responsibilities

| Name                      | Role/Responsibilities                               |
|---------------------------|-----------------------------------------------------|
| Leisha Khapre             | Leader (liaison)/Speaker/Quality Assurance/Reviewer |
| Maddie Bird, Inayaa Khoja | Project Management/Analyst/Documenter/Speaker       |
| Jade Winebright           | Documenter/Reviewer/Developer/Speaker               |
| Gaurav Shatoor            | Developer/Designer/Researcher/Speaker               |

### 4. Communication

- We will respond to messages within 3 hours.
- Primary communication method: Text Message.
- Weekly check-in: Wednesday 9am.
- Shared folder: <https://drive.google.com/drive/folders/0AMnO4fpIqhI9Uk9PVA>

### 5. Attendance & Deadlines

- Attend scheduled meetings unless prior notice is given.
- Submit work by agreed deadlines.
- Notify the team early if delays are expected.

### 6. Conflict Resolution

- Discuss issues directly and very respectfully.
- If unresolved, escalate to the course instructor and TA.

## 7. Commitment

By signing below, each member agrees to uphold this agreement and contribute equally to the success of the project.

## 8. Punishment

Service to the group.

### Signatures:

Inayaa Khoja      Date: 1/18/26

Jade Winebright      Date: 1/18/26

Maddie Bird      Date: 1/18/26

Leisha Khapre      Date: 1/18/26

Gaurav Shatoor      Date: 1/18/26

**Question 4:** How did you and your team members collectively establish goals and plan tasks?

We established goals early on through our project planning and then broke them down into smaller tasks based on the main parts of the system, such as data generation, preprocessing, detection, and evaluation. Tasks were assigned based on team groups and individual strengths, and we tracked progress through regular check-ins and deadlines. This made it easier to stay on schedule and adjust when needed.

**Question 5:** Why do you think that you and your team members have met the project objectives?

We met the project objectives because our final solution directly addresses the client's main needs of improving safety, increasing efficiency, and creating a reproducible workflow. By developing an automated anomaly detection pipeline, we reduced reliance on manual diver-based inspections, which supports safer operations. The system also improves efficiency by processing and analyzing point cloud data more quickly and consistently. In addition, our workflow is fully documented and reproducible, allowing it to be reused or expanded in the future, which was another key need identified by the client. See *1.4. Client Needs* on pg. 7.

### 8.1.6 Student Outcome 6

*An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.*

**Question 1:** Briefly describe how your team developed and conducted experimentation.

The team developed the experimentation process by utilizing Blender for synthetic data generation. We configured multiple maritime scenes and utilized the BIAnder sensor suite to simulate a boat-mounted LiDAR sensor. This involved iterative experimentation with random object placement and scanner parameter calibration to ensure high-fidelity data.

For the detection process, we implemented a PointNet++ architecture as suggested by Shaukat, Kückenmeister, Bader, and Kirste (2025). Through hyperparameter tuning and testing, we determined that an ensemble of models performing anomaly detection in stages yielded the most accurate results. See 3.2 *Solution Concepts* on pg. 15

**Question 2:** Briefly describe how your team analyzed and interpreted the collected data.

Once the synthetic data was generated, we utilized the Open3D library in Python for 3D visualization. We performed visual inspections of the point clouds to ensure that the bounding box annotations and labels for maritime objects were accurate and consistent. To interpret the performance of our models, we analyzed the collected output using statistical metrics, specifically Precision, Recall, and F1 scores. We used confusion matrices to identify specific classes where the model struggled, allowing us to interpret which environmental factors most significantly impacted detection accuracy. See 3.4 *Selected Solution* on pg. 17

**Question 3:** Briefly describe how, and based on what engineering concepts, your team drew its conclusions.

We conducted a comparative cost analysis between traditional manual diver inspections and the proposed Project LiAISon solution. By evaluating metrics such as daily operating costs, personnel requirements, and insurance liability, we concluded that the automated workflow could eliminate over \$20,000 in startup field-trial costs and increase inspection throughput by 80%

To determine the effectiveness of our detection models, we utilized a variety of statistical performance metrics, including precision, recall, F1 score, and confusion matrices. These metrics allowed us to quantitatively conclude that the PointNet model provided the optimal balance of accuracy and computational efficiency compared to other alternatives. We also employed a traceability matrix to link high-level client needs (such as safety and automation) directly to specific engineering actions and technical objectives. See 3.4 *Selected Solution* on pg. 17

### 8.1.7 Student Outcome 7

*An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.*

**Question 1:** How did your team acquire knowledge about the potential solutions for the problem at hand?

During the project charter development phase, the team acquired knowledge by conducting a comprehensive literature review on 3D object detection and maritime anomaly identification. We researched various machine learning architectures—including SASA and PointNet—to evaluate their computational efficiency and accuracy in processing high-density point clouds.

Additionally, we consulted our advisors, Dr. Alfredo Garcia and Anshul Yadav. Dr. Garcia specifically recommended exploring autoencoders to reduce data dimensionality while preserving critical features, which informed our early analysis of alternatives. See *1.6 Literature Review* on pg.12 and *6 References* on pg.25.

**Question 2:** What method(s) and learning strategies were adopted by your team to further learn about the project and its potential solutions?

Our team deepened its understanding of the project through a combination of regular meetings with advisors, collaborative problem-solving sessions, and independent technical research. We met consistently with Dr. Garcia and Anshul to clarify concepts, validate assumptions, and refine our modeling direction. Team members also engaged in independent reading of academic papers and documentation on point-cloud processing and synthetic data generation, then shared insights during internal working sessions. This blend of expert guidance, peer collaboration, and self-directed learning allowed us to build the knowledge needed to evaluate and develop effective solutions. See *1.6 Literature Review* on pg.12 and sample advisor communications from *8.1.3 Student Outcome 3 (Question 2)* on pg. 30

**Question 3:** How did your team apply new knowledge to develop potential solutions for the problem at hand?

We applied our newly acquired knowledge directly to the design and implementation of the anomaly-detection pipeline. Insights from the literature review guided our selection of PointNet as the final model due to its robustness to unordered point sets and noise. Advisor recommendations informed our early exploration of autoencoders and dimensionality-reduction techniques, which helped us evaluate trade-offs between reconstruction accuracy and computational cost. Knowledge gained from experimentation shaped our preprocessing pipeline, including seafloor removal, DBSCAN clustering, and normalization strategies. Finally, lessons from peer collaboration and expert feedback helped refine our evaluation metrics and improve the interpretability of our results.

## 8.2 ABET Criterion 5d: Curriculum Capstone Requirements

*The curriculum must include a culminating major engineering design experience that 1) incorporates appropriate engineering standards and multiple constraints, and 2) is based on the knowledge and skills acquired in earlier course work.*

### 8.2.1 Engineering Standards

ISO 19115: Geographic Information Metadata. This standard was used to structure metadata for point clouds, including scene parameters, anomaly labels, and coordinate frames. See 4.5.2 *Engineering Standards* on pg. 23.

ISO/IEC 25012: Data Quality Model. This standard guided our requirements for accuracy, consistency, and completeness in synthetic datasets and preprocessing. See 4.5.2 *Engineering Standards* on pg. 23.

PEP 8 – Python Code Style Standard. This standard ensured maintainable, readable, and consistent code across preprocessing, detection, and evaluation scripts. See 4.5.2 *Engineering Standards* on pg. 23.

### 8.2.2 Multiple Realistic Constraints

Technical Constraints: Limited access to high-performance computing resources and model accuracy limitations required lightweight architectures and efficient preprocessing. See 4.5.1 *Client Constraints* on pg. 23.

Time Constraints: A fixed academic semester timeline restricted the number of scenes, model iterations, and testing cycles we could complete. See 4.5.1 *Client Constraints* on pg. 23.

Economic Constraints: With no project budget, we relied entirely on free tools such as Blender, Google Colab, and open-source libraries, all with runtime and memory limits. See 4.5.1 *Client Constraints* on pg. 23.

Legal & Regulatory Constraints: We ensured compliance with data privacy and avoided use of proprietary sonar/LiDAR data. See 4.5.1 *Client Constraints* on pg. 23.

### 8.2.3 Use of Knowledge and Skills Acquired in Earlier Course Work

Table 5. Previous DAEN Courses Referenced during Capstone Project

| Phase |                                                                                                | DAEN Course(s) Used          | Rationale                                                                                                                                                                                                |
|-------|------------------------------------------------------------------------------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Information gathering<br>Customer needs<br>Technical objectives                                | DAEN 459, DAEN 460           | Learned to identify client needs before starting the project and establishing technical objectives to fulfill said needs.<br><i>1 Phase 1: Discover on page 7</i><br><i>2 Phase 2: Define on page 12</i> |
| 2     | As-is concept of operations<br>System requirements<br>Engineering actions                      | DAEN 459, DAEN 460           | Introduced to CONOPS and traceability matrix.<br><i>2 Phase 2: Define on page 12</i><br><i>3.1 To-Be Concept of Operations on page 15</i>                                                                |
| 3     | To-be concept of operations<br>Solution concept generation<br>Analysis of alternatives         | DAEN 323                     | Learned about the different kinds of models and their use cases.<br><i>3.2 Solution Concepts on page 15</i><br><i>3.3 Analysis of Alternatives on page 16</i><br><i>3.4 Selected Solution on page 17</i> |
| 4     | Final design architecture<br>Cost, performance, and risk analyses<br>Constraints and standards | ISEN 302; DAEN 331; DAEN 323 | Learned economic analysis of engineering projects; DAEN courses helped in comparing computational costs in phase 4<br><i>4 Phase 4: Deliver on page 17</i>                                               |

## Final Deliverables Checklist

|                                     |                                                                                                                                                                                                                          |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> | Our team competed in the <b>Engineering Virtual Project Showcase</b> .                                                                                                                                                   |
| <input checked="" type="checkbox"/> | Our team submitted the “ <b>Phase 4 Presentation</b> ” on Canvas.                                                                                                                                                        |
| <input checked="" type="checkbox"/> | Our team submitted the “ <b>Final Report + ABET Documentation</b> ” on Canvas.                                                                                                                                           |
| <input checked="" type="checkbox"/> | Our team submitted all “ <b>Supplemental Data/Documents</b> ” as a .zip file on Canvas.                                                                                                                                  |
| <input checked="" type="checkbox"/> | Our team emailed the following documents to our sponsor ( <b>with TAs cc’d</b> ):<br>“ <b>Phase 4 Presentation</b> ”<br>“ <b>Final Report + ABET Documentation</b> ”<br>“ <b>Supplemental Data/Documents</b> ” .zip file |